Syed Ali Abbas Abedi, Ph.D.

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SUMMARY

Computational chemist with a Ph.D. and 6+ years of hands-on experience in quantum chemical simulations, reaction mechanism analysis, and machine learning applied to molecular property prediction. Skilled in developing hybrid computational workflows that combine DFT/TD-DFT calculations with Python-based data pipelines and ML models to assess chemical reactivity, stability, and degradation pathways. Published 23 papers (488 citations, h-index 11) in journals including *Nature Protocols*, *Nature Methods*, and *JACS*, with a consistent track record of translating computational predictions into actionable recommendations for experimental teams across 15+ interdisciplinary collaborations.

EDUCATION

Ph.D., Science, Mathematics and Technology (Computational Chemistry) 2020 to 2024
Singapore University of Technology and Design (SUTD), Singapore
Thesis: Computational Design and Mechanistic Investigation of Fluorophores Based on Excited-State Conformational Dynamics

M.S., Mathematics 2017 to 2019
Lahore University of Management Sciences (LUMS), Pakistan

B.Sc., Chemical Engineering 2012 to 2016
University of Engineering and Technology, Lahore, Pakistan

TECHNICAL SKILLS

Quantum Chemistry: DFT, TD-DFT, CASSCF, CASPT2, spin-orbit coupling calculations, conical intersection optimization, reaction pathway mapping, transition state searches, thermodynamic barrier estimation, solvation modelling (PCM/CPCM)

Machine Learning: Scikit-learn (regression, classification, random forests, gradient boosting), feature engineering from molecular descriptors, model validation (cross-validation, error metrics), experience with PyTorch for neural network prototyping

Software: Gaussian, ORCA, Quantum ESPRESSO, Multiwfn, RDKit, VMD

Programming and Automation: Python (Pandas, NumPy, SciPy, Matplotlib), Bash/Shell scripting, MATLAB, Git, Linux command-line, HPC workflow automation (SLURM, PBS)

Data Analysis: Statistical analysis of heterogeneous computed and experimental datasets, automated log-file parsing, data visualization, trend identification across large compound libraries

RESEARCH EXPERIENCE

Research Fellow Jan 2026 to Present
Nanyang Technological University (NTU), School of Chemistry, Chemical Engineering and Biotechnology, Singapore

- Developed a hybrid quantum chemistry and ML workflow to screen 200+ molecular candidates for photostability and reactivity, ranking compounds by computed degradation susceptibility to prioritize experimental testing.
- Built end-to-end Python pipelines that extract 40+ molecular descriptors from Gaussian output files, feed them into Scikit-learn regression models, and generate ranked candidate lists, cutting manual analysis time by 40%.
- Investigated non-radiative decay pathways and spin-orbit coupling mechanisms to identify structural features that control molecular stability, directly informing experimental design decisions.
- Collaborated with experimental teams across 4 countries, translating computational predictions into synthesis recommendations that contributed to 5 publications in 2025 and 2026.

Research Fellow (Postdoctoral) Sep 2024 to Jan 2026
Singapore University of Technology and Design (SUTD), Singapore

- Conducted computational studies to predict photothermal degradation pathways, correlating DFT-computed energy barriers with experimental failure modes across 50+ compounds with 85%+ agreement.
- Benchmarked 12 DFT functionals and 8 basis sets with solvation models (PCM/CPCM) to establish an optimal accuracy-vs-cost protocol for reactivity assessments, reducing per-job compute time by 30%.
- Automated parsing and statistical analysis of 10,000+ quantum chemistry log files using Bash/Python, enabling rapid identification of stability trends across heterogeneous datasets.
- Published a first-author computational protocol in *Nature Protocols* for diagnosing photoinduced electron transfer mechanisms, providing a validated and reproducible workflow now adopted by external groups.

Visiting Researcher Apr 2023 to Jul 2023

University College London (UCL) and Queen Mary University of London, UK

- Collaborated with experimental scientists to validate multi-reference CASSCF models for reactive potential energy surfaces involving conical intersections, contributing to 2 joint publications.
- Resolved persistent convergence issues in high-level excited-state calculations, enabling reliable mechanistic modelling of chemical degradation pathways.

Graduate Research Assistant Aug 2020 to Jun 2024

Singapore University of Technology and Design (SUTD), Singapore

- Developed and validated computational protocols for predicting molecular reactivity and conformational stability using DFT/TD-DFT, applied across 12 co-authored studies on structurally diverse compound families.
- Built reusable Python toolkits for automated quantum chemistry job submission, output parsing, and property extraction on HPC clusters, adopted by 8 lab members.
- Worked with heterogeneous experimental and computed datasets to establish structure-property relationships linking molecular design to functional outcomes.

SELECTED PUBLICATIONS

23 peer-reviewed papers | 488 citations | h-index: 11

1. **Abedi** et al. (2025). A Computational Protocol for Uncovering Photoinduced Electron Transfer Mechanisms. *Nature Protocols* (accepted). [First author; validated workflow]
2. Kim et al., **Abedi** et al. (2025). Super-Photostable Organic Dye for Single-Protein Imaging. *Nature Methods*, 22, 550 to 558. [Stability prediction for experimental design]
3. Sosnin, **Abedi** et al. (2025). Clicked Hydrazone Photoswitches. *J. Am. Chem. Soc.*, 147, 14930 to 14935. [Reactivity modelling]
4. Yan et al., **Abedi** et al. (2021). Fluorescence Umpolung for Sensing. *Nature Commun.*, 12, 3869. [Mechanism analysis]
5. **Abedi**, Chi et al. (2021). Restriction of TICT Enables Aggregation-Induced Emission. *J. Phys. Chem. A*, 125, 8397 to 8403. [First author; conformational reactivity]

Additional publications in Angew. Chem. Int. Ed. (4), Adv. Sci., Adv. Funct. Mater., CCS Chemistry, and others. Full list available upon request.

ADDITIONAL INFORMATION

- Available to relocate to Beerse, Belgium. Eligible for EU work visa sponsorship.
- Comfortable working in hybrid and interdisciplinary team environments. Experienced in liaising with experimental scientists to define validation strategies and translate model outputs into actionable recommendations.
- Vice President, LUMS Alumni Chapter Singapore. First-generation academic.